



Optimization via the strategic law of large numbers

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This paper proposes a framework for the global optimization of a possibly multimodal continuous function in a bounded rectangular domain. We first show that global optimization is equivalent to an optimal (sampling) strategy formation in a two-armed decision model with known distributions, based on the strategic law of large numbers we establish. There are many optimal strategies in general. We show that a concrete strategy using the sign of the partial gradient of the unique solution to a parabolic partial differential equation (PDE) is asymptotically optimal. Motivated by these results, we propose a class of Strategic Monte Carlo Optimization (SMCO) algorithms, which uses a simple strategy that makes coordinate-wise two-armed decisions based on the signs of the partial gradient (or practically the first difference) of the objective function, without the need of solving PDEs. Under some sufficient conditions, we establish that our SMCO algorithm converges to a local optimizer from a single starting point, and to a global optimizer under a growing set of starting points. Extensive numerical studies demonstrate the suitability of our SMCO algorithms for global optimization well beyond the theoretical guarantees established herein. For a wide range of deterministic and random test functions with challenging landscapes (multimodal, nondifferentiable, discontinuous), our SMCO algorithms perform robustly well, even in high-dimensional ($d = 200 \sim 1,000$) settings. In fact, our algorithms outperform many state-of-the-art global optimizers, as well as local algorithms (with the same set of starting points as ours).

black box global optimization | strategic limit theorem | two-armed machine | strategic sampling | high dimension

It has been a notoriously difficult task to develop fast algorithms for global optimization of high-dimensional and potentially multimodal continuous functions, which is known as the “curse of optimality” phenomenon. In this paper, we propose an approach to solving the following optimization problem

$$\max_{x \in \Gamma} f(x), \quad [1]$$

where $\Gamma := [\underline{\mu}_1, \bar{\mu}_1] \times \cdots \times [\underline{\mu}_d, \bar{\mu}_d] \subset \mathbb{R}^d$, and $f: \Gamma \rightarrow \mathbb{R}$ is a continuous function.

Problem Eq. 1 is well defined in the sense that there exists a global maximizer of any continuous deterministic function f over a compact region $\Gamma \subset \mathbb{R}^d$. Under mild regularity conditions, it is known that a continuous random function also has a global maximizer on a compact region Γ almost surely without assuming f being a global concave function [see, for example, Cox (1)]. However, when f is a multimodal and high-dimensional continuous function, such as Rastrigin function with $d = 200, 400, 500, 1,000$, it is very difficult to develop a fast algorithm to solve Problem Eq. 1 and to mathematically prove its convergence to a global maximizer. Various Stochastic Gradient Descent (SGD) algorithms rely on accurate estimation of gradients (some also rely on Hessians) for iterative updates and are the key behind recent success of deep learning (neural networks). The gradient based algorithms are only shown to converge to a global maximum when the objective function f is strictly concave, however. There are also many popular meta-heuristic global optimization algorithms for Problem Eq. 1, such as Simulated Annealing (SA) of Kirkpatrick et al. (2) and the Particle Swarm Optimization (PSO) of Eberhart and Kennedy (3). SA has been shown to converge weakly to global optimal points [Geman and Hwang (4)] but its convergence rate is known to be slow and it performs poorly in practice. Although popular in applied work, the PSO has not been mathematically proved to converge to a global maximum for a general multimodal continuous function yet.

We propose a framework for the global optimization problem Eq. 1. Our core theoretical results, *Theorems 2.1 and 2.2*, establish that solving problem Eq. 1 is equivalent

Significance

This work proposes a framework for global optimization, showing that global optimization is equivalent to optimal strategy formation in a two-armed decision problem with known distributions, based on the strategic law of large numbers we establish. This paradigm shift simplifies the action space for optimization problems, and motivates us to propose a class of Strategic Monte Carlo Optimization (SMCO) algorithms that feature automatic dynamic step size adaptation. Numerical studies demonstrate the fast and robust convergence of SMCO algorithms to global maximizer(s) for multidimensional, multimodal functions with challenging optimization landscapes. We anticipate that this work will advance the theoretical understanding of the connection between classic bandit process and global optimization problems, and inspire additional global optimization algorithms in our framework.

The authors declare no competing interest.

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to the search for an optimal (sampling) strategy in a two-armed decision model with known distributions, based on the strategic law of large numbers (StLLN). Precisely, the StLLN we establish provides a theoretical framework that translates the global optimization problem Eq. 1 into an equivalent problem of finding an approximately optimal strategy in a two-armed bandit model with known distributions. Both StLLN and the classical LLN concern the asymptotic behavior of the sample mean, but StLLN considers strategic sampling from two distributions, while the classical LLN focuses on (nonstrategic) sampling from a single distribution. Hence, the classical LLN can be viewed as a special case of StLLN in which the two distributions coincide.

There are in general many approximately optimal strategies to solve problem Eq. 1. Our *Theorem 2.3* further establishes that one approximately optimal strategy is to use the signs of partial gradient of the unique solution to a parabolic PDE. For low-dimensional (e.g., $d = 1, 2$) problems, one could apply finite-difference method to numerically solve the PDE and then design a two-armed sampling algorithm according to the sign of the partial gradient of the numerical solution of the PDE. Unfortunately, solving a high-dimensional PDE numerically is in general very difficult. Interestingly, under symmetry and some smoothness condition on a one-dimensional f , the sign of gradient of the unique solution to the PDE coincides with the sign of the gradient of the original f itself (*Remark 2.2*).

Motivated by the above results, we propose a simple optimization algorithm based on a tractable strategy for the two-armed decision model, using only the sign information of the partial derivative of the original function f (instead of the PDE solution). The running sample mean under iterative application of this strategy is then returned as the candidate optimizer by our algorithm. We call this the “Strategic Monte Carlo Optimization” (SMCO) algorithm. Although this simple strategy may not coincide with the asymptotic optimal strategy using the sign of the gradient of the PDE solution in multidimensional problems, our *Theorem 2.4* shows that the SMCO algorithm is nevertheless guaranteed to converge to local optimizers. We show in *Theorem 2.5* that the SMCO algorithm, when augmented with a growing set of starting points, converges to a global optimizer. In addition, when f has a unique local maximizer x^* on Γ , *Theorem 2.6* establishes a convergence rate of our simple strategic sample mean to x^* . In practice, we implement the SMCO algorithms with multiple randomly generated starting points, and use step-adaptive finite differences in place of the true gradients. We also provide two slightly modified versions of the basic SMCO algorithm, SMCO-R and SMCO-BR, with bookkeeping of running maximum and simple additional programming enhancements. The source code of SMCO/-R/-BR is publicly available on GitHub.

We conduct a wide range of numerical experiments to analyze the performance of our SMCO algorithms in solving complex optimization problems in comparison with existing popular optimizers in R. We consider various multimodal deterministic test functions widely used as benchmarks in the global optimization literature, such as Rastrigin, Ackely, Michalewicz, Griewank, and more, of dimensions ranging from 1 to 1,000. In addition, we also consider a variety of random test functions with challenging optimization landscapes (some nondifferentiable and even discontinuous), such as sample criterion function for maximum score estimation of Manski (5), empirical welfare function for optimal treatment assignment, conditional moment inequalities in dynamic binary choice models, in-sample approximation, and out-of-sample prediction based on ReLU neural networks

in regression and classification settings, and Cauchy likelihood functions. We compare our SMCO algorithms with a range of widely used state-of-the-art optimization algorithms/packages implemented in R, including six local algorithms (GD, SignGD, ADAM, SPSA, L-BFGS, and BOBYQA) and six global ones (GenSA, SA, DEoptim, STOGO, GA, and PSO).

Our SMCO algorithms, designed using the simple sign-based strategy, perform remarkably well in complex global optimization problems, to an extent that goes well beyond the theoretical guarantees we establish. Specifically, our SMCO algorithms are consistently able to solve almost all test optimization problems to a reasonable degree of accuracy, using only a very limited number of randomly generated starting points for dimensions up to $d = 1,000$. In contrast, all other algorithms considered here, with the sole exception of GenSA (an improved version of SA), suffer from drastic errors in at least a number of test cases. In particular, the six local optimizers perform significantly worse than ours in many test cases, even though they use the same set of starting points as ours. This provides strong evidence that the performance of our SMCO algorithms in global optimization cannot be explained by the use of multiple starting points alone. In addition, our SMCO algorithms compare favorably to most of the global optimizers, especially so for higher dimensional test functions. This is again remarkable given that our SMCO algorithms only use a small number of starting points even in higher dimensions. Although GenSA sometimes performs slightly better than ours, it is extremely slow as dimension increases, which explains why GenSA is rarely used in modern high dimensional optimization problems. The performance of our SMCO algorithms thus appears unexpectedly strong, especially given that all of these competing optimizers have been widely tested and improved over many years, and that there are likely to be many aspects of the SMCO algorithms and our simple R code that can be significantly improved (such as coding in C++).

The numerical experiments demonstrate the promise of our SMCO algorithms, and more generally the optimization framework we proposed based on StLLN with two-armed decision strategy. We note that the SMCO algorithms are based on a simple sign-based strategy only, although robust to multimodal landscape and noisy gradient, they may not be as fast as some other algorithms when f is strictly concave and differentiable. This suggests an array of future research questions under our two-armed StLLN framework: to analyze the performances of other strategies under this framework; to search for and/or implement other (approximately) optimal strategies in a tractable way; or to provide additional theoretical results that articulate the observed good performance of our SMCO algorithms in solving complex global optimization problems.

1.1. Related Literature.

- (i) Decision model-driven optimization: Recently, bandit process, a simplified decision model of reinforcement learning, has shown significant success in optimization by employing random strategy with exploration, because this exploration expands the search space, enhances flexibility, and supports learning through trial and error. See refs. 6–9 and the references therein for various applications of bandit algorithms with unknown arm distributions in optimization. Nevertheless, our work formally establishes the equivalence of global optimization problem Eq. 1 to the optimal policy/strategy formation in a two-armed decision model with known arm distributions. The StLLN for the constructed decision model (*Theorem 2.1*) is our theoretical foundation for recasting

the global optimization of any continuous function as a decision rule problem, which is inspired by the LLN under the nonlinear expectation of Peng (10). We further propose a simple SMCO algorithm with multistart and rigorously establish its convergence to the global maximizer(s).

- (ii) Two-armed bandit problems: Our optimization framework is reminiscent of, but in fact distinct from, the classical two-armed bandit problems. While traditional bandit problems focus on maximizing cumulative rewards under uncertainty about arm distributions (11–14), we search for global maximizer(s) of a possibly nonconcave continuous function f over Γ according to a two-armed decision model with known arm distributions. Precisely, SMCO selects between two fixed arms (or $2 \times d$ for d -dimensional problems) based on the sign changes of the objective function f 's coordinate-wise partial gradient, ensuring quicker convergence to maximum points while reverting to standard Monte Carlo sampling when the arms coincide.
- (iii) Gradient Descent and its variants: Numerous optimization algorithms are based on gradient descent (GD) and stochastic gradient descent (SGD), including adaptive gradient (15–18) and sign gradient descent (SignGD) (19–21). Traditional GD and SGD are known to be sensitive to the choice of starting points, the updating step-sizes, and the accuracy of gradient estimates for its updates. Guo and Fu (9) proposed to apply multiarm bandit to select the starting points for SGD and demonstrated the convergence advantages numerically. Adaptive GD methods use past stochastic gradients to adjust step-sizes, which potentially lead to faster convergence and mitigate gradient explosion. However, these methods still face challenges, particularly in high-dimensional settings, where step size determination remains complex. Additionally, they are prone to gradient explosion in early training phases (22, 23), causing slow convergence or divergence. SignGD uses the gradient's sign for updates and is shown to improve the stability of the traditional GD in some cases [Li et al. (24)], but its performance remains highly sensitive to the choices of starting points and updating step-sizes. Our SMCO may look like SignGD in that both use sign information for updates and hence are robust to noisy gradients but is actually quite different. Specifically, our SMCO is motivated from the StLLN, and its sample mean formulation automatically adapts step sizes with respect to both the iteration counter and the current optimizer position, with randomness incorporated in the mean-zero known distributions around the upper/lower domain bounds. Our SMCO relies on two-armed decision rule-generated samples and is less sensitive to the choice of starting points. Indeed, extensive simulations demonstrate that our algorithm compares very favorably with GD, SignGD, and its variants in terms of optimization performances, especially in high-dimensional optimization problems with challenging landscapes (multimodal, nondifferentiable, discontinuous).

1.2. Outline of the Rest of the Paper. Section 2 presents the strategic law of large numbers, the corresponding optimization framework, our SMCO algorithm design, and theoretical guarantees on the convergence of the SMCO algorithm. Section 3 conducts a wide range of numerical comparisons with the existing popular optimization algorithms. We conclude with Section 4. *SI Appendix* contains consisting of all the proofs, technical lemmas, the details on the practical implementations of the SMCO algorithms, and additional numerical experiments using deterministic and random functions.

2. Theory and Algorithms

2.1. New Framework: Optimization via Strategic Law of Large Numbers. We use a simple decision model to illustrate our approach to the optimization problem Eq. 1. Suppose that there are d two-armed decision models: These d decision models are independent of each other and differ only in the mean value corresponding to their arms. For $j = 1, \dots, d$, machine j has two arms with populations X_j and Y_j and different known means $\bar{\mu}_j$ and $\underline{\mu}_j$ (without loss of generality, we assume $\bar{\mu}_j > \underline{\mu}_j$). One can draw one d -dimensional sample from these d two-armed decision models at each round; that is, the d components of the sample come from the d two-armed decision models, and decision model j can be drawn from either arm X_j or arm Y_j , but not both. If X_j and Y_j are identical, the decision model j has a single arm and it becomes the classical Monte Carlo draws.

We will focus on the global optimization of a continuous function f on a bounded rectangular region Γ , $\Gamma := [\underline{\mu}_1, \bar{\mu}_1] \times [\underline{\mu}_2, \bar{\mu}_2] \times \dots \times [\underline{\mu}_d, \bar{\mu}_d] \subset \mathbb{R}^d$. For a small positive δ , we assume that the function f is well defined on the δ -extension region Γ_δ of Γ defined by $\Gamma_\delta := [\underline{\mu}_1 - \delta, \bar{\mu}_1 + \delta] \times [\underline{\mu}_2 - \delta, \bar{\mu}_2 + \delta] \times \dots \times [\underline{\mu}_d - \delta, \bar{\mu}_d + \delta]$. In this paper all random variables are defined on a given probability space $(\Omega, \mathcal{F}, P)$. Consider d two-armed decision models, for $1 \leq j \leq d$, let (X_j, Y_j) denote the random reward of one of these d two-armed decision models. For simplicity, we assume that $\{X_j : 1 \leq j \leq d\}$ and $\{Y_j : 1 \leq j \leq d\}$ satisfy

$$X_j = \bar{\mu}_j + \xi_j, \quad Y_j = \underline{\mu}_j + \xi'_j, \quad [2]$$

where ξ_j is a bounded random variable with mean 0, variance σ^2 and bounded by $\delta > 0$ (that is, $|\xi_j| \leq \delta$), ξ'_j is an independent copy of ξ_j . For all $1 \leq j \leq d$, let $\{X_{ij} : i \geq 1\}$ and $\{Y_{ij} : i \geq 1\}$ respectively denote the outcomes from the arm X_j and arm Y_j at the i th round. We define a sampling strategy θ as a sequence of (binary) random vectors $\theta := \{\theta_1, \dots, \theta_n, \dots\}$, where $\theta_i := (\theta_{i1}, \dots, \theta_{id}) \in \{0, 1\}^d$ and $\theta_{ij} = 1$ (respectively, $\theta_{ij} = 0$) means in the i th round we pull the arm X_j (respectively, Y_j) of the j th slot machine. Let $\{Z_i^\theta := (Z_{i1}^\theta, \dots, Z_{id}^\theta) : i \geq 1\}$ be a sequence of d -dimensional random vectors denoting the i th reward obtained from these d slot machines related to the strategy θ . That is, for $j = 1, \dots, d$,

$$Z_{ij}^\theta := \theta_{ij}X_{ij} + (1 - \theta_{ij})Y_{ij} = \begin{cases} X_{ij}, & \text{if } \theta_{ij} = 1, \\ Y_{ij}, & \text{if } \theta_{ij} = 0. \end{cases} \quad [3]$$

We call a sampling strategy $\theta = \{\theta_1, \dots, \theta_n, \dots\}$ *admissible* if θ_n is $\mathcal{H}_{n-1}^\theta$ -measurable for all $n \geq 1$, where

$$\mathcal{H}_{n-1}^\theta = \sigma\{\eta, Z_1^\theta, \dots, Z_{n-1}^\theta, \theta_1, \dots, \theta_{n-1}\} \text{ for } n > 1, \mathcal{H}_0^\theta = \sigma\{\eta\},$$

where $\eta := (\eta_1, \dots, \eta_d) \in \Gamma$ denotes the initial point, which could be nonrandom or randomly drawn from a known distribution (such as uniform distribution on the domain or a deterministic grid distribution on the hyperdiagonal of Γ). Let Θ denote the collection of all admissible sampling strategies. For any strategy $\theta \in \Theta$, we let

$$S_0^\theta = \eta, \quad \text{and for all } n \geq 1, \quad S_n^\theta = S_{n-1}^\theta + Z_n^\theta. \quad [4]$$

Theorem 2.1 (Strategic Law of Large Numbers and a Two-sided Framework for Optimization). *Let the d two-armed decision models $\{(X_j, Y_j) : 1 \leq j \leq d\}$ satisfy Eq. 2. Let Z_n^θ and S_n^θ*

be defined in Eqs. 3 and 4. Let f be any continuous function defined on Γ_δ . Then, the global optimization problem Eq. 1 is equivalent to the problem of asymptotically maximizing the “reward” with f

$$\max_{x \in \Gamma} f(x) = \lim_{n \rightarrow \infty} \sup_{\theta \in \Theta} E \left[f \left(\frac{S_n^\theta}{n} \right) \right]. \quad [5]$$

Furthermore, we have,

$$\sup_{\theta \in \Theta} P \left(\lim_{n \rightarrow \infty} \sup_{x \in \Gamma} f \left(\frac{S_n^\theta}{n} \right) = \max_{x \in \Gamma} f(x) \right) = 1 \quad [6]$$

According to Theorem 2.1, the global optimization Eq. 1 can be transformed into two-armed decision model problem, which can be equivalently expressed as

$$\lim_{n \rightarrow \infty} \inf_{\theta \in \Theta} R_f^\theta(n) = 0, \quad [7]$$

where $R_f^\theta(n) := \max_{x \in \Gamma} f(x) - E \left[f \left(\frac{S_n^\theta}{n} \right) \right]$ denote the “averaged regret” with f under θ . Therefore, to solve the global optimization problem Eq. 1, one just needs to find an asymptotically optimal (sampling) strategy $\theta^o \in \Theta$ such that $\lim_{n \rightarrow \infty} R_f^{\theta^o}(n) = 0$. In addition, if the objective function f is Lipschitz continuous, we can obtain the following speed of convergence result.

Theorem 2.2 (Convergence Rate). Under conditions of Theorem 2.1, if f is Lipschitz continuous on Γ_δ (with a Lipschitz constant L_f), then:

$$\left| \sup_{\theta \in \Theta} E \left[f \left(\frac{S_n^\theta}{n} \right) \right] - \max_{x \in \Gamma} f(x) \right| \leq L_f \left(\frac{\sigma \sqrt{d} + \sum_{j=1}^d \frac{|\bar{\mu}_j - \mu_j|}{2}}{\sqrt{n}} + \frac{\sqrt{E[\|\eta\|^2]}}{n} \right). \quad [8]$$

$$[9]$$

Remark 2.1. In our decision model, when these two arms (populations) coincide, our strategic laws of large numbers (StLLN) (Theorems 2.1 and 2.2) degenerates into the classical LLN, which serves as the theoretical foundation for the classical Monte Carlo simulation method. Consequently, we have the following viewpoints: Repeatedly operating a single-armed decision model corresponds to the conclusions drawn from the classical laws of large numbers and can thus be utilized for simulating the integration of a function. However, strategically operating a two-armed decision model aligns with our proposed strategic laws of large numbers, enabling its application in simulating the extremum of functions.

Theorems 2.1 and 2.2 establish that any optimization problem Eq. 1 can be solved by finding some optimal (sampling) strategy of the two-armed decision model.* Both results are significant and broadly applicable, and motivate us to consider a class of optimization algorithms that compute sample means over draws from one of two prespecified distributions around the two boundary points (in each dimension), along with a potentially optimal strategy to choose from the two distributions.

2.2. A Concrete Asymptotically Optimal Strategy. In the next theorem, we will suggest an asymptotically optimal (sampling) strategy using the signs of the partial gradients of the solutions to some parabolic PDEs associated with the Lipschitz continuous function f . This PDE approach enables us to find the global optima of function f on region Γ via a two-armed decision model.

For any small enough $\epsilon > 0$, consider the following parabolic PDE

$$\begin{cases} \partial_t u_\epsilon(t, x) + \sup_{p \in \Gamma} [D_x u_\epsilon(t, x) \cdot p] + \frac{\epsilon^2}{2} \text{tr} [D_x^2 u_\epsilon(t, x)] = 0, \\ u_\epsilon(1 + \epsilon, x) = \hat{f}(x), \end{cases} \quad [10]$$

for $(t, x) \in [0, 1 + \epsilon) \times \mathbb{R}^d$, where $\hat{f}$ is an extension of f defined on $\mathbb{R}^d$, $D_x u_\epsilon$ and $D_x^2 u_\epsilon$ denote the gradient vector and the Hessian matrix of the function $x \rightarrow u_\epsilon(t, x)$ respectively. Under the bounded Lipschitz continuity assumption on $\hat{f}$, this PDE Eq. 10 is actually the HJB equation, and has a unique solution u_ϵ [see Yong and Zhou (27, Theorem 5.2, Ch. 4)]. Moreover, the solution $u_\epsilon(t, x)$ is continuously differentiable in x [see Peng (10, Theorem C.4.5), also see SI Appendix, Appendix A for details]. We can establish the following refinement of Theorem 2.2.

Theorem 2.3. Let f be a Lipschitz continuous function on Γ_δ with Lipschitz constant L_f , and $\hat{f}$ be a bounded Lipschitz continuous (with Lipschitz constant L_f) extension of f defined on $\mathbb{R}^d$. For any small $\epsilon > 0$ let u_ϵ be the unique solution of PDE Eq. 10. For any $n \in \mathbb{N}$, define the strategy $\hat{\theta}^{n, \epsilon} = (\hat{\theta}_1^{n, \epsilon}, \dots, \hat{\theta}_i^{n, \epsilon}, \dots)$ with $\hat{\theta}_i^{n, \epsilon} = (\hat{\theta}_{i,1}^{n, \epsilon}, \dots, \hat{\theta}_{i,d}^{n, \epsilon})$ as follows, for $j = 1, \dots, d$,

$$\hat{\theta}_{i,j}^{n, \epsilon} = \begin{cases} 1, & \text{if } \partial_{x_j} u_\epsilon \left(\frac{i}{n}, \frac{S_{i-1}^{\hat{\theta}^{n, \epsilon}}}{n} \right) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad [11]$$

for $1 \leq i \leq n$, and $\hat{\theta}_{i,j}^{n, \epsilon} = 1$ for $i > n$. Then: $\{\hat{\theta}^{n, \epsilon}\}$ is asymptotically ϵ -optimal in the sense that

$$\left| E \left[f \left(\frac{S_n^{\hat{\theta}^{n, \epsilon}}}{n} \right) \right] - \max_{x \in \Gamma} f(x) \right| \leq L_f \epsilon (2C_\Gamma + \sqrt{\epsilon} + \sqrt{1 + \epsilon}) + \frac{C_1 E[\|\eta\|]}{n} + C \left(\frac{1}{n} \sum_{i=1}^d \left(\sigma^2 + \frac{(\bar{\mu}_i - \mu_i)^2}{4} \right) + \frac{\sum_{i=1}^d |\bar{\mu}_i - \mu_i|}{2n} + 2\epsilon \sqrt{\frac{d}{n}} + \epsilon^2 \right), \quad [12]$$

$$[13]$$

where $C_\Gamma > 0$ is a finite constant depending on the boundary of Γ , and $C, C_1 > 0$ are finite constants depending on f .

According to Theorem 2.3, we can use the sign of the gradient vector $D_x u_\epsilon$ to construct a sequence of strategies $\hat{\theta}^{n, \epsilon}$ such that $f(S_n^{\hat{\theta}^{n, \epsilon}}/n)$ approximates $\max_{x \in \Gamma} f(x)$.

Remark 2.2. To construct a sequence of asymptotically ϵ -optimal strategies, we only need to know the sign of each component of the gradient vector $D_x u_\epsilon$. When f and its extension $\hat{f}$ satisfy some assumptions, the sign of the gradient vector $D_x u_\epsilon$ can be denoted by the sign of the gradient f . For example,

*For other recent studies on nonlinear and strategic limit theory for decision model problems, the reader is referred to refs. 25 and 26 and references therein.

- When $\hat{f} \in C^3(\mathbb{R})$, assume that for any $t \in [0, 1]$, the nodal set of $\partial_x u_\epsilon(t, x)$ is denoted by $D = \{d_k, k = 0, \pm 1, \pm 2, \dots\}$. Moreover, if $\hat{f}$ is strictly monotone on $[d_k, d_{k+1}]$, and the derivatives $\hat{f}^{(k)}$ ($k = 0, 1, 2, 3$) have at most polynomial growth, then (see more details in refs. 28 and 29):

$$\left\{ \partial_x u_\epsilon \left(\frac{i}{n}, \frac{S_{i-1}^{\hat{\theta}^{n,\epsilon}}}{n} \right) > 0 \right\} \Leftrightarrow \left\{ \partial_x \hat{f} \left(\frac{S_{i-1}^{\hat{\theta}^{n,\epsilon}}}{n} \right) > 0 \right\}.$$

Hence the optimal strategy defined in Eq. 11 can be replaced by the sign of the gradient vector $D_x \hat{f}$. A sufficient condition for the above assumption to hold is that $\hat{f} \in C^3(\mathbb{R})$ is a symmetric function with symmetric points $D = \{d_k, k = 0, \pm 1, \pm 2, \dots\}$ and strictly monotone on $[d_k, d_{k+1}]$.

It is well known that obtaining an analytical solution for PDE Eq. 10 is generally intractable, often necessitating numerical methods to approximate the sign of its solution's gradient. Consequently, the operational feasibility of locating the global optimum of function f via the aforementioned approach faces limitations. However, Remark 2.2 shows that for certain specific functions f , the sign of the gradient of the PDE solution coincides with that of the gradient of f itself. This motivates us to investigate whether we can directly leverage the sign of function f 's gradient to construct a class of asymptotically optimal strategies. Such strategies would locate the optimum of f within the two-armed decision-making framework. To address this, we present a simple algorithm, which we call "Strategic Monte Carlo Optimization" in the next subsection.

2.3. Strategic Monte Carlo Optimization (SMCO) Algorithm. Motivated by the theoretical results in previous subsections, we now introduce a simple strategy θ^* that just uses the sign information of the coordinate-wise partial derivative of f .

At round $m \geq 1$, we construct $\theta_m^* = (\theta_{m,1}^*, \dots, \theta_{m,d}^*)$ as follows:[†]

$$\theta_{m,j}^* = \begin{cases} 1, & \text{if } \partial_{x_j} f \left(\frac{S_{m-1}^{\theta_{m-1}^*}}{m-1} \right) \geq 0; \\ 0, & \text{otherwise,} \end{cases} \quad j = 1, \dots, d. \quad [14]$$

Intuitively, suppose the current sample mean is in a region where the function f is increasing, then we aim to maximize f by selecting a sample from the arm with a higher population mean in the next round; if the sample mean is in a region where the function f is decreasing, then we select the next sample from the arm with a lower (population) mean. SI Appendix, Fig. S1C illustrates the different iterative mechanisms for attaining the maximum value by the GD method (blue arrow line) and our SMCO (red arrow line) respectively. It also demonstrates that under the simple sign-based strategy θ^* given in Eq. 14, the sample average $S_n^{\theta^*}/n$ converges to the one of the local maxima of f . This result constitutes the strategic law of large numbers for the decision model problem (Theorem 2.4 below). In the one-dimensional setting, our proposed two-armed decision model process devises strategies that combine the two arms with known distributions to identify the maximum value of a nonconcave function f . This approach differs from the classical two-armed decision model, which explores optimal arms to achieve the maximum average reward for each arm under a monotonic function f with unknown distributions.

[†]When $m = 1$, $S_0^{\theta^*}/0$ can be defined as η , which is uniformly distributed on Γ .

Algorithm 1 Strategic Monte Carlo Optimization (SMCO)

Input: Function f , tolerance level ϵ , initial point $\hat{x}_0$, with $S_0^{\theta^*} := \hat{x}_0$.

for $n = 0$ **to** ∞ **do**

for $j = 1$ **to** d **do**

if $\partial_{x_j} f(\hat{x}_n) \geq 0$ **then**

$\theta_{n+1,j}^* = 1$, draw $Z_{n+1,j}^{\theta^*} = X_{n+1,j}$ from arm X_j (larger mean in Eq. 2);

else

$\theta_{n+1,j}^* = 0$, draw $Z_{n+1,j}^{\theta^*} = Y_{n+1,j}$ from arm Y_j (smaller mean in Eq. 2).

end

end

Update $S_{n+1}^{\theta^*} := S_n^{\theta^*} + Z_{n+1}^{\theta^*}$, with

$Z_{n+1}^{\theta^*} = (Z_{n+1,1}^{\theta^*}, \dots, Z_{n+1,d}^{\theta^*})$.

Update $\hat{x}_{n+1} = S_{n+1}^{\theta^*}/(n+1)$.

if $|f(\hat{x}_{n+1}) - f(\hat{x}_n)| \leq \epsilon$ **then**

break

end

end

Output: Final point $\hat{x}^* := \hat{x}_{n+1}$ and final function value $\hat{f}^* := f(\hat{x}^*)$.

The following theorem establishes the StLLN according to the sign-based strategy θ^* described in Algorithm 1, which ensures that under θ^* , the sample mean $S_n^{\theta^*}/n$ converges to one of the local maxima of f on Γ .

Theorem 2.4 (Local Optimization under the Sign-Based Strategy). Let f be a continuously differentiable function on Γ_δ , and satisfy the following condition: For any $k = 1, \dots, d$ and a fixed integer $K_k \in \mathbb{N}$, there exist two sets of points $\underline{\mu}_k \leq c_k^1 < \dots < c_k^{K_k} \leq \bar{\mu}_k$ and $\underline{\mu}_k - \delta = b_k^0 < b_k^1 < \dots < b_k^{K_k-1} < b_k^{K_k} = \bar{\mu}_k + \delta$, such that for any fixed $x = (x_1, \dots, x_d) \in \Gamma_\delta$, the function $f(x_1, \dots, x_{k-1}, \cdot, x_{k+1}, \dots, x_d)$ is strictly increasing on (b_k^{j-1}, c_k^j) and strictly decreasing on (c_k^j, b_k^j) for $j = 1, \dots, K_k$.

Let $\mathcal{M}$ denote all local maximum points off on Γ , and for any $\epsilon > 0$, $\mathcal{M}_\epsilon = \bigcup_{c \in \mathcal{M}} B(c, \epsilon)$, where $B(c, \epsilon) = \{x \in \Gamma_\delta : \|x - c\|_\infty < \epsilon\}$. Let $\theta^* = (\theta_1^*, \dots, \theta_m^*, \dots)$, with $\theta_m^* = (\theta_{m,1}^*, \dots, \theta_{m,d}^*)$, be the strategy defined in Eq. 14. Then:

$$\lim_{n \rightarrow \infty} P \left(\frac{S_n^{\theta^*}}{n} \in \mathcal{M}_\epsilon \right) = 1. \quad [15]$$

Remark 2.3. In Theorem 2.4, we make a sufficient condition that f is continuously differentiable. In fact, the above conclusion can be obtained similarly for continuous function f that is not differentiable. For a nondifferentiable function, we can approximate it by a smooth function, for example, $f_\epsilon(x) = \int f(x + \epsilon y) \phi(y) dy \rightarrow f(x)$, as $\epsilon \rightarrow 0$, where ϕ is the probability density function of a standard normal distribution. Then, one can simulate the extremes of f by applying the method to f_ϵ for a small enough ϵ .

Remark 2.4. The coordinate separability condition essentially ensures that (local) coordinate-wise updates are sufficient to

increase the value of the function in each iteration. The condition plays a role that is akin to the coordinate-wise smoothness condition (Lipchitz property of gradients) imposed for sign gradient descent (19) and coordinate gradient descent (30). Note that we do not impose additional smoothness conditions beyond the stated monotonicity requirements.

Next, we show that SMCO, when augmented with a growing set of multiple starting points, attains the global maximum for functions with multiple local maxima. Specifically, we show theoretically that, given a starting point in a local neighborhood of the global maximizer, the SMCO algorithm modified with a “boosted” starting iteration index is guaranteed to converge to the global maximizer.

Theorem 2.5 (Global Convergence). *Let f be a continuously differentiable function on Γ_δ . For any global maximum x^* of f on Γ , assume that there exists $\delta_0 > 0$, such that for any $y \in B(x^*, \delta_0)$, $\text{sgn}(\partial_{x_i} f(y)) = \text{sgn}(x_i^* - y_i)$, $i = 1, \dots, d$. Then there exists an integer $N_0 > 0$ such that for any $m \geq N_0$, whenever $S_m^\theta/m \in B(x^*, \delta_0)$, it holds that for any $\varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} P\left(\left\|\frac{S_n^{\theta'}}{n} - x^*\right\|_\infty < \varepsilon\right) = 1, \quad [16]$$

where $B(x^*, \delta_0) = \{x \in \Gamma_\delta : \|x - x^*\|_\infty < \delta_0\}$ and $\theta' = (\theta_1, \dots, \theta_m, \theta_{m+1}^*, \theta_{m+2}^*, \dots)$.

Consequently, given a growing (deterministically, randomly, or quasi-randomly generated) set of starting points $\mathcal{X}_m$ such that $\#(\mathcal{X}_m) = m$ and $P(B(x^*, \delta_0) \cap \mathcal{X}_m \neq \emptyset) \rightarrow 1$ as $m \rightarrow \infty$, the SMCO algorithm run from all points in $\mathcal{X}_m$ with a “boosted” starting iteration index[‡] $n_0 = N_0$ is guaranteed to attain a global maximizer x^* asymptotically as $n, m \rightarrow \infty$.

Finally, we present a rate of convergence of our SMCO algorithm 1 when f has a unique local maximum point $x^* \in \Gamma$, which could be on the boundary of Γ .

Theorem 2.6 (Convergence Rate II). *Let f be a continuous twice-differential function on Γ_δ , and assume that f has a unique local maximum point $x^* = (x_1^*, \dots, x_d^*)$ on Γ , and further the i th component of f is strictly increasing on $(\underline{\mu}_i - \delta, x_i^*)$ and strictly decreasing on $(x_i^*, \bar{\mu}_i + \delta)$. Then:*

$$\begin{aligned} & \left| E \left[f \left(\frac{S_n^{\theta^*}}{n} \right) \right] - f(x^*) \right| \\ & \leq \max \left\{ L_f \left(\frac{\sigma \sqrt{d}}{\sqrt{n}} + \frac{\sqrt{E[\|\eta\|^2]}}{n} \right), \right. \\ & \quad \left. \frac{K_f \left(d\sigma^2 + \sum_{j=1}^d (\bar{\mu}_j - \underline{\mu}_j)^2 \right) + L_f E[\|\eta\|]}{n} \right\}, \end{aligned}$$

where L_f is the Lipschitz constant of f , $K_f = \max_{i,j} \sup_{x \in \Gamma_\delta} |\partial_{x_i x_j}^2 f(x)|$ and σ^2 is the uniform variances of $\{X_j\}$ and $\{Y_j\}$ as defined in Eq. 2. Moreover,

$$E \left[\left\| \frac{S_n^{\theta^*}}{n} - x^* \right\|^2 \right] \leq \frac{d\sigma^2 + \sum_{j=1}^d (\bar{\mu}_j - \underline{\mu}_j)^2}{n} + \frac{E[\|\eta\|^2]}{n^2}.$$

[‡]Formally, running SMCO with a boosted iteration index n_0 means the following: Given a starting point x_0 , set the initial sum $S_0 := n_0 x_0$, and in the n -th iteration, update the running sum as in Algorithm 1 but compute the sample mean $x_{(n+1)} := S_{(n+1)} / (n + n_0)$. Effectively, this is as if the algorithm starts with the iteration counter $n = n_0$ instead of $n = 0$.

3. Numerical Experiments

3.1. Three Implemented Versions of SMCO Algorithms. We now provide a brief description of how we practically implement Algorithm 1. See SI Appendix, Appendix B for additional details.

3.1.1. Finite differencing with adaptive step size. Theoretically, Algorithm 1 requires the computation of the gradients. In practice, we compute the signs of finite differences in lieu of the signs of gradients. Specifically, at iteration n , we define the step size in the j -th coordinate $h_{n,j} := (\bar{\mu}_j - \underline{\mu}_j) / (n + 1)$ and compute the finite difference $\Delta_{x_j} f(\hat{x}_n) := f(\hat{x}_n + h_{n,j} \cdot e_j) - f(\hat{x}_n - h_{n,j} \cdot e_j)$, where e_j denotes the j -th elementary vector. We then use the sign information $\Delta_{x_j} f(\hat{x}_n) \geq 0$ in place of $\partial_{x_j} f(\hat{x}_n) \geq 0$ in Algorithm 1. Observe that $\Delta_{x_j} f(x) \geq 0$ is equivalent to $\Delta_{x_j} f(x) / (2h_{n,j}) \geq 0$, and that $\Delta_{x_j} f(x) / (2h_{n,j}) \rightarrow \partial_{x_j} f(x)$ as $n \rightarrow \infty$. Hence, the finite difference calculation above is a valid approximation of the gradient, establishing the coherence of our practical implementation with the gradient-based theoretical algorithm. It is worth noting that we set the step size of the finite-difference adaptively as $h_n \sim 1/(n + 1)$ instead of a small constant (say, 10^{-8}) as commonly used. This is also motivated by our StLLN-based algorithm, under which the running maximizer $\hat{x}_n$ takes the form of a sample mean, which naturally implies that the position updating step size from $\hat{x}_n$ to $\hat{x}_{n+1}$ is of the order of $1/(n + 1)$. Since the updating of $\hat{x}_n$ is exactly guided by the (sign) information in the finite difference calculation, it is thus heuristically natural to align the finite-difference step size h_n with the position updating step size in their orders of magnitude.

Based on the above, we implement the following three versions of our SMCO algorithms, which we use in the numerical experiments in Section 3.

- (1) SMCO: Our baseline algorithm that implements Algorithm 1 using the adaptive-size finite differencing as described above. We set the random variable ξ_j in Eq. 2 to be $U[-\delta_j, \delta_j]$ with $\delta_j := \delta^* (\bar{\mu}_j - \underline{\mu}_j)$, and $\delta^* = 0.05$ as the default bound expansion parameter. We set $\varepsilon = 1e - 6$ as the default tolerance level and 500 as the default maximum number of iterations, the initial point $\hat{x}_0 = \eta$ can vary with test function dimension d .
- (2) SMCO-R: A refined version of SMCO with two programming/meta-heuristic enhancements. First, SMCO-R records and exploits the running maximum of all test function evaluations: This is purely bookkeeping and does not add any additional evaluations of the test functions beyond those for finite-difference calculations and convergence checks. From a programming perspective, the running maximum generated in past iterations is “free information” that requires essentially no additional computation and memory storage. Second, SMCO-R adopts a two-stage procedure: It starts by running SMCO for at most 50% iterations of the designated maximum number of iterations, and then runs the rest 50% iterations with a more local “refined” search (and smaller step sizes). Specifically, the end point of the initial stage is used as the start point of the second stage, namely $x_0^{(2)}$, and the second stage SMCO with the starting iteration counter set to $n_0 = 1,000$. Specifically, the partial sum is initialized at $S_0^{(2)} := 1000x_0^{(2)}$, the next point is updated according to $x_1^{(2)} := (S_0^{(2)} + Z_1^{(2)}) / 1001$, and etc. Other hyperparameters are set to be the same as in SMCO.

(3) SMCO-BR: A “boosted” version of SMCO-R that runs two separate passes of SMCO-R with: i) the first pass is exactly the same as SMCO-R, ii) the second pass runs a more local version of SMCO-R with the starting iteration counter set to $n_0 = 100$,[§] as motivated by *Theorem 2.5*. Specifically, given any starting point x_0 , the second pass initializes the running sum as $S_0 := 100x_0$, computes the first update as $x_1 = (S_0 + Z_1)/101$, and so on and so forth. Since SMCO-BR effectively runs two separate passes of SMCO-R, to make the comparison fair, we set the maximum number of iterations to be half of that for SMCO/SMCO-R, so that the running times of the three versions are similar. Other hyperparameters are set to be the same as in SMCO.

3.2. Alternative Optimization Algorithms in R for Comparison.

Below we list two groups of state-of-the-art optimization algorithms (implemented in R) to which we compare our SMCO-R/-BR algorithms in terms of performance accuracy (in root mean squared error) and computation speed (in second). See *Appendix A*, *Appendix C* for more information on these algorithms and their hyperparameter configurations.

3.2.1. Algorithm group I: Six local optimizers. This set consists of six gradient-based or derivative-free optimization algorithms that are designed to converge to a local optimum starting from a given starting point: 1) GD, 2) L-BFGS, 3) SignGD, 4) SPSA, 5) ADAM, and 6) BOBYQA. To make the comparison as fair as possible, we pass the exact sets of starting points used for our SMCO algorithms to the following algorithms as well: For single-start comparisons, our SMCO algorithms and the Group I algorithms are all run from a common random starting point; for multistart comparisons, they are all run from the same set of starting points.

3.2.2. Algorithm group II: Six global optimizers. This set consists of six R packages that implement meta-heuristic global optimization algorithms: 1) GenSA, 2) SA, 3) DEoptim, 4) STOGO, 5) GA, and 6) PSO. We use the default hyperparameters set in these packages. Since these algorithms are explicitly designed for global optimization (and many already implement multiple starting points from within), we do not pass any generated starting points to these algorithms whenever applicable.[¶]

3.3. Optimization of Deterministic Test Functions. We first run the optimization algorithms on the following set of deterministic test functions, which have been widely used to benchmark the performances of global optimization algorithms. The function definitions (and hyperparameter values, if applicable) follow those given in the optimization section of the Virtual Library of Simulated Experiments: Test Functions and Datasets[#] (31). We have tested a wide array of test functions from different categories with various dimensionalities. Due to page limit, we only report results in the main text on the following four test functions: 1) Rastrigin function on $[-5.12, 5.12]^d$. This function has a unique global minimum of value 0 at the origin and multiple global maxima at $x \in \{\pm 4.52299\}^d$. 2) Ackley function on $[-32.768, 32.768]^d$. This function has a

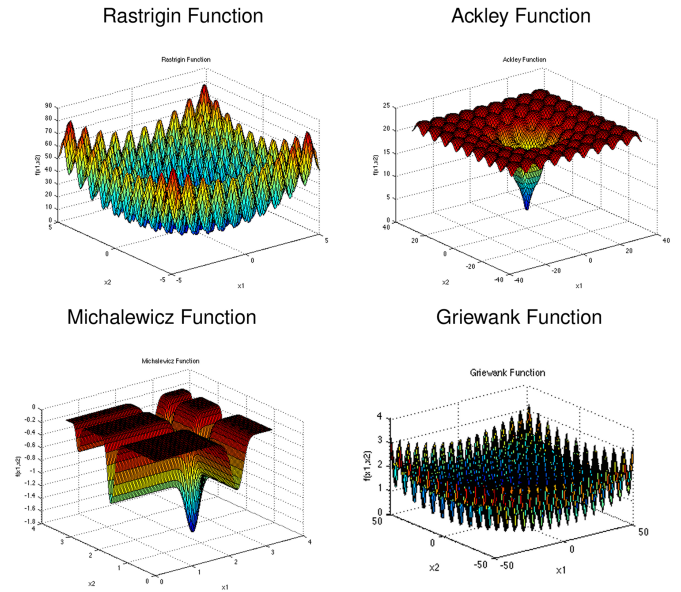


Fig. 1. Visualization of test functions. Retrieved from: <https://www.sfu.ca/~ssurjano/optimization.html>.

unique global minimum of value 0 at the origin and multiple global maxima of unknown values. 3) Griewank function on $[-600, 600]^d$. This function has a unique global minimum of value 0 at the origin and multiple global maxima of unknown values. 4) Michalewicz function on $[0, \pi]^d$. This function has a unique global minimum (of values reported up to $d = 75$) and multiple global maxima of the value 0 at the vertices $\{0, \pi\}^d$. These functions are constructed to have challenging optimization landscapes with many local optima, boundary optima, (near) flat regions/directions, steep drops/ridges, or a combination of these. Fig. 1 contains visualizations (retrieved from the Virtual Library of Simulated Experiments) of these test functions in 2-dimensional forms.

One notable feature of several of the test functions above lies in their symmetric forms, i.e., they are constructed with symmetric domains and its global minimum attained at the origin (or, the center point of the domain). Such a kind of symmetry is unlikely to be found in realistic optimization problems. This would make the optimization problem drastically easier for any algorithm that explicitly or implicitly searches the center point of the domain, making the comparison between algorithms less clear and fair. Hence, we randomly rotate, shift, and asymmetricize the default domains of the test functions above, which we formally describe in the following. Specifically, for a given test function f on its default domain $\prod_{j=1}^d [\underline{\mu}_j, \bar{\mu}_j]$, write $\Delta\mu_j := \bar{\mu}_j - \underline{\mu}_j$ be the length of the j -th coordinate domain. Then, we shift and asymmetricize the domain in each coordinate independently as follows: For each $j = 1, \dots, d$, we draw $r_j \sim \text{Bernoulli}(0.5)$, $v_j \sim N(0, 1)$, and $v_j \sim U[0, 1]$, and set

$$\begin{aligned}\underline{\mu}_j^{mod} &:= \underline{\mu}_j + [v_j + r_j(0.2 + 0.1v_j) - (1 - r_j)(0.4 + 0.2v_j)] \Delta\mu_j, \\ \bar{\mu}_j^{mod} &:= \bar{\mu}_j + [v_j + r_j(0.4 + 0.2v_j) - (1 - r_j)(0.2 + 0.1v_j)] \Delta\mu_j.\end{aligned}$$

Essentially, the above functions constitute the following operations in each coordinate. (i) Randomly shift the center point of the original domain by $v_j \cdot \Delta\mu_j$; (ii) Randomly decide whether to asymmetricize the domain rightward ($r_j = 1$) or leftward

[§]Note that this starting iteration counter is set separately and differently from the one in the second stage of SMCO-R. We set $n_0 = 100$ given that it works fairly well and some preliminary trials of larger numbers do not seem to change the results much. Thus we did not further tune this parameter for potential performance gains.

[¶]A few of these global algorithms nevertheless require a user-designated starting point, in which case we pass a start point randomly chosen from the set of multiple starting points generated for our SMCO and the Group I algorithms.

[#]Available online at <https://www.sfu.ca/~ssurjano/optimization.html>.

($r_j = 0$). If asymmetrizing rightward, i.e., $r_j = 1$, then push the lower bound $\underline{\mu}_j$ rightward by $25\% \pm 5\%$ of $\Delta\mu_j$, and push the upper bound $\bar{\mu}_j$ rightward by $50\% \pm 10\%$ of $\Delta\mu_j$. If asymmetrizing leftward, i.e., $r_j = 0$, then push the lower bound $\underline{\mu}_j$ leftward by $50\% \pm 10\%$ of $\Delta\mu_j$, and push the upper bound $\bar{\mu}_j$ leftward by $25\% \pm 5\%$ of $\Delta\mu_j$. Note that the domain of each coordinate is perturbed independently, so the “asymmetrization directions” generally differ across coordinates. After shifting and asymmetrizing the domain into $\mathcal{X}^{mod} := \prod_{j=1}^d [\underline{\mu}_j^{mod}, \bar{\mu}_j^{mod}]$ as described above, we further randomly rotate the domain by randomly drawing a rotation (orthonormal) matrix Q of dimension d , and define the final test function by $f^{mod}(x) := f(Q(x - v \odot \Delta\mu))$, where $\odot$ represents the coordinate-wise (Hadamard) product.

For each of the test functions (with rotation and domain modification), we test the optimization algorithms both for minimization and maximization. Due to the randomness in the random starting point generation as well as the stochastic nature

of many optimization algorithms used, we run multiple replications of all optimization algorithms and report summary statistics for their performance across the replications. Specifically, we report: “Time,” “RMSE,” and “AE_{xx},” whose definitions we now explain. “Time” reports the average time (in second) it takes for a particular algorithm to return the final function value, averaged across all the replications. “RMSE” and “AE_{xx}” are relative accuracy measures defined as follows. Since the true maximum and minimum values may be different on the rotated, enlarged, and asymmetrized domain from those on the default domains, we first define “Best Value” as the best value found by all algorithms (ours and those in Groups I & II) over all replications. We then define “RMSE” as the square root of the mean squared error between the value returned by a specific algorithm and the “Best Value” found, and define “AE_{xx}” as the xx-th percentile of the absolute error between the found value and the “Best Value.”

Table 1 summarizes how our SMCO algorithms Group I (local) algorithms in maximizing and minimizing all four test functions of dimension $d = 2$, when run from a single

Table 1. Deterministic functions, dimension $d = 2$: Comparison with Group I algorithms

Performance		RMSE	AE50	AE95	AE99	Time	RMSE	AE50	AE95	AE99	Time
Min	Function: Best value	Rastrigin: 0				Griewank: 0					
	SMCO	7.8	8.0	13.0	13.0	0.016	9.9	9.8	11.1	11.6	0.017
	SMCO-R	2.8	2.0	5.0	8.0	0.016	0.04	0.03	0.08	0.12	0.017
	SMCO-BR	2.7	2.0	5.0	8.0	0.016	0.06	0.03	0.11	0.16	0.018
	GD	36.7	17.0	80.9	104.7	0.010	196.2	141.4	345.6	437.4	0.032
	SignGD	62.3	42.2	115.4	156.2	0.035	208.4	141.4	403.3	492.5	0.026
	SPSA	39.6	16.9	84.6	135.4	0.002	137.9	72.6	277.6	330.7	0.003
	ADAM	15.4	8.3	30.0	48.5	0.008	211.5	142.1	418.3	506.2	0.014
	L-BFGS	43.8	15.9	97.5	115.5	0.0004	141.9	78.8	289.6	333.8	0.0005
	BOBYQA	42.9	9.9	88.5	135.4	0.002	18.9	0.2	0.8	15.9	0.002
	Function: Best value	Ackley: 4.44e-16				Michalewicz: -1.801303					
	SMCO	3.1	0.4	0.9	19.8	0.018	1.25	1.15	1.64	1.64	0.018
	SMCO-R	2.3	0.01	0.01	19.5	0.019	1.11	1.14	1.37	1.43	0.018
	SMCO-BR	2.6	0.01	0.01	19.4	0.018	1.09	1.14	1.15	1.37	0.018
	GD	19.0	19.9	20.0	20.3	0.021	1.74	1.84	1.99	1.99	0.012
	SignGD	19.0	19.9	20.0	20.0	0.034	1.55	1.63	1.99	1.99	0.023
	SPSA	19.0	19.9	20.0	20.0	0.002	1.45	1.34	1.99	1.99	0.004
	ADAM	19.1	19.9	20.0	20.7	0.001	1.84	1.99	2.00	2.00	0.006
	L-BFGS	18.8	19.9	20.0	20.4	0.0005	1.65	1.63	1.99	1.99	0.001
	BOBYQA	15.1	18.0	20.0	20.2	0.002	1.47	1.37	1.99	1.99	0.003
Max	Function: Best value	Rastrigin: 213.5824				Griewank: 622.3434					
	SMCO	17.4	10.3	10.4	88.7	0.015	2.5	1.0	5.5	6.7	0.016
	SMCO-R	13.4	10.2	10.2	58.5	0.015	1.1	3e-13	2.1	3.6	0.015
	SMCO-BR	15.6	10.2	10.2	80.6	0.015	2.2	0.1	4.8	8.5	0.016
	GD	85.3	82.7	173.3	192.9	0.004	469.6	477.0	609.1	618.5	0.033
	SignGD	130.2	130.9	171.1	173.1	0.035	471.1	477.0	609.1	618.5	0.027
	SPSA	85.1	80.6	167.2	176.0	0.004	472.0	477.6	609.1	618.9	0.013
	L-BFGS	82.8	81.9	165.0	171.1	0.0004	447.6	459.9	608.7	618.5	0.0005
	BOBYQA	73.6	20.3	167.0	171.1	0.002	204.4	0.0	293.4	575.1	0.002
	Function: Best value	Ackley: 22.35029				Michalewicz: 1.21406					
	SMCO	0.20	0.18	0.29	0.37	0.017	0.25	0.21	0.41	0.41	0.017
	SMCO-R	0.03	0.02	0.06	0.08	0.018	0.17	6.5e-7	0.41	0.41	0.016
	SMCO-BR	0.02	0.01	0.05	0.08	0.018	0.15	7.6e-7	0.41	0.41	0.016
	GD	1.90	0.08	4.44	8.83	0.017	0.85	0.72	1.21	1.21	0.011
	SignGD	2.20	0.08	4.75	9.55	0.032	0.65	0.41	1.21	1.21	0.024
	SPSA	2.19	0.08	4.75	9.55	0.002	0.94	1.21	1.21	1.21	0.002
	L-BFGS	1.66	0.08	3.64	7.71	0.000	0.77	0.41	1.21	1.21	0.001
	BOBYQA	0.25	0.02	0.61	1.19	0.002	0.58	0.41	1.21	1.21	0.002

Based on 500 reps. In each rep, all algorithms run from a single uniformly drawn starting point. Computed using Posit Cloud. SMCO-R/-BR run with 1e-6 tolerance and 500 maximum iterations.

uniformly generated starting point, based on 500 replications. First, our SMCO algorithms tend to produce significantly more accurate results than all Group I algorithms in most of the test configurations reported in Table 1. The RMSE and AE percentiles of our SMCO algorithms are often orders of magnitude smaller than those of other algorithms in most configurations. Second, only our SMCO algorithms display robust performance across all eight test configurations in comparison with all other algorithm. Finally, our SMCO algorithms display remarkably stable performance in terms of the error percentiles: The differences between the 50th, 95th, and 99th error percentiles of our SMCO algorithms tend to be substantially smaller than those of all other Group I algorithms, demonstrating that the performance of our SMCO algorithms is much *less sensitive* to the random starting point relative to others.

In *SI Appendix, Tables S2–S5* report comparisons of our SMCO algorithms with both Group I and II algorithms for dimensions $d = 2, 10, 20, 50$, in which our SMCO algorithms and all Group I algorithms are run with $10\sqrt{d} \approx 14, 32, 45$, and 71 uniformly drawn starting points. The general pattern there shows that: First, our SMCO algorithms perform robustly well across most settings, both in terms of RMSE and AE99. Second, Group I (local) algorithms perform significantly worse than SMCO and Group II (global) algorithms, especially when the dimension increases. Consequently, due to the space limit, we only compare our SMCO algorithms with Group II algorithms in higher-dimensional settings in the main text.

We next report results on how our SMCO algorithms compare with Group II (global) algorithms in a high-dimensional ($d = 200$) setting. Here our SMCO algorithms are run from $\sqrt{d} \approx 14$ uniformly drawn starting points in a parallelized manner. The Group II (global) algorithms are run without any specified starting point (when the specification of a starting point is not required by the algorithm) or one randomly drawn starting point (when the specification of a starting point is required by the algorithm), again at their default settings. Again, our (multi-start) SMCO algorithms demonstrate robust performance across almost all test configurations, with speed and accuracy levels that compare competitively, and in fact often favorably, to Group II

(global) algorithms. Among Group II (global) algorithms, GenSA appears to be the only one that shows overall better performance than our SMCO algorithms in Table 2. That said, it still yields noticeably larger errors (in terms of RMSE and AE99) than ours, say, in the minimization of Rastrigin and Ackley functions. Additional results in higher dimensions ($d = 400, 500, 1,000$), as well as various sensitivity checks with respect to the hyperparameters of SMCO/-R/-BR, are available in *SI Appendix, Appendix D*.

3.4. Optimization of Random Functions. In this section, we investigate two randomly generated test functions that arise from two important economic contexts. The first test function is the sample criterion for maximum score estimation, as introduced by Manski (5). The maximum score estimator can be used to estimate model parameters in discrete choice models under weak assumptions on the error terms, and since (5, 32) there has been much subsequent work in economics that either extends the applicability of the maximum score estimation approach or analyzes its intriguing properties. Specifically, the sample criterion function for maximum score estimator is defined as $f_{MS}(\beta) := \frac{1}{N} \sum_{i=1}^N Y_i \mathbb{1}\{X_{i1} + X'_{i,-1}\beta \geq 0\}$ where $(X_i, Y_i)_{i=1}^N$ is a random sample with $X_i = (X_{i1}, X_{i,-1}) \in \mathbb{R}^{d+1}$ being a vector of covariates and Y_i being a binary outcome variable generated by the model $Y_i = \mathbb{1}\{X_{i1} + X'_{i,-1}\beta_0 + \epsilon_i \geq 0\}$ under the conditional median restriction $\text{med}(\epsilon_i | X_i) = 0$ at the true parameter β_0 . The maximum score estimator is then defined as any solution to the maximization problem $\max_{\beta} f_{MS}(\beta)$. Note that the parameter to be optimized over, β , enters through the discontinuous indicator function, and furthermore the outcome variable Y_i is also a binary variable itself. Due to the discreteness issues, the maximization of the maximum score criterion function is well known to be computationally challenging, especially when the dimension of parameters becomes (moderately) high. We investigate the performance of our SMCO algorithms in optimizing the maximum score criterion function. Specifically, we build the test function f_{MS} using simulated data generated as follows. We simulate $X_{i1} \sim \mathcal{N}(0, d)$ and $X_{i2} \sim \dots \sim X_{i,d+1} \sim \mathcal{N}(0, 1)$. We set $\beta_0 \in \mathbb{R}^d$ through random draws of $\beta_{0,j} \sim \mathcal{N}(0, 1)$. We

Table 2. Deterministic functions, dimension $d = 200$: Comparison with Group II algorithms

Test function		Rastrigin			Griewank			Ackley			Michalewicz		
Performance		RMSE	AE99	Time	RMSE	AE99	Time	RMSE	AE99	Time	RMSE	AE99	Time
Min	Best value	1,596.819			2.35e-13			19.85938			-62.77543		
	SMCOms	285	375	34.9	1.07	1.07	34.4	0.101	0.106	31.9	42.22	44.08	42.7
	SMCOms-R	262	308	33.4	0.95	0.95	37.2	0.009	0.014	32.3	28.42	29.41	42.5
	SMCOms-BR	272	324	32.6	0.96	0.96	36.5	0.010	0.015	32.6	30.33	32.21	43.0
	GenSA	1,718	2,011	268.3	3.05e-11	6.10e-11	302.2	0.107	0.114	271.7	6.27	12.03	813.4
	SA	3,947	4,338	0.2	11,626	14,002	0.1	1.651	1.679	0.3	52.76	54.53	0.36
	DEoptim	3,813	3,968	31.5	5,856	6,086	33.4	1.545	1.563	32.2	47.41	48.91	43.5
	STOGO	1,293	1,293	285.6	4.38e-09	4.38e-09	307.1	0	0	295.4	43.48	43.48	404.4
	GA	2,496	2,623	0.5	7,372	7,920	0.5	1.450	1.518	0.5	49.28	52.30	0.61
	PSO	590	812	4.9	81	143	5.1	1.546	1.590	5.1	43.13	45.36	6.19
Max	Best value	18,239.57			72,331.4			22.33939			69.53472		
	SMCOms	7,328	7,648	34.7	192	199	38.0	0.119	0.125	31.6	36.67	38.25	42.48
	SMCOms-R	4,419	4,939	33.3	278	287	36.2	0.105	0.110	30.8	20.91	22.16	42.47
	SMCOms-BR	4,512	5,043	34.7	601	627	37.4	0.030	0.034	31.7	25.43	26.87	42.49
	GenSA	2,605	2,975	275.9	0	0	312.7	0.004	0.007	273.5	4.65	8.98	814.1
	SA	5,653	7,180	0.3	32,221	36,635	0.3	0.488	0.505	0.3	55.48	57.14	0.37
	DEoptim	7,978	8,191	31.8	36,012	36,465	33.7	0.421	0.432	32.6	49.83	50.98	43.8
	STOGO	10,987	10,987	290.0	9,908	9,908	310.0	0.120	0.120	299.9	45.26	45.26	407.8
	GA	8,977	9,310	0.5	45,978	46,953	0.5	0.450	0.489	0.5	49.86	51.20	0.61
	PSO	998	1,719	5.0	20,792	23,940	5.2	0.429	0.448	5.2	46.50	48.88	6.23

Based on 10 reps. SMCO/-R/-BR run from $\sqrt{d} \approx 14$ uniformly drawn starting points in parallel. Run on high-performance cluster using 15 cores on 1 node.

Table 3. Maximization of random functions, dimension $d = 5, 10, 20$, & 50

Dimension		$d = 5$			$d = 10$			$d = 20$			$d = 50$		
Performance		RMSE	AE99	Time	RMSE	AE99	Time	RMSE	AE99	Time	RMSE	AE99	Time
Maximum score estimation: $\max_{\beta} f_{MS}(\beta)$													
Best value		0.42			0.448			0.462			0.424		
(I)	GD	0.135	0.196	0.06	0.146	0.190	0.03	0.167	0.198	0.08	0.157	0.174	0.13
	SignGD	0.135	0.196	0.08	0.146	0.190	0.03	0.167	0.198	0.09	0.157	0.174	0.11
	ADAM	0.135	0.196	0.07	0.146	0.190	0.06	0.167	0.198	0.08	0.157	0.174	0.11
	SPSA	0.135	0.196	0.11	0.146	0.190	0.08	0.167	0.198	0.11	0.157	0.174	0.12
	L-BFGS	0.135	0.196	0.07	0.146	0.190	0.07	0.167	0.198	0.09	0.157	0.174	0.13
	BOBYQA	0.084	0.116	0.07	0.100	0.141	0.08	0.130	0.153	0.09	0.134	0.155	0.22
	SMCO	0.008	0.012	0.20	0.009	0.012	0.33	0.013	0.019	0.76	0.056	0.068	5.09
	SMCO-R	0.002	0.004	0.21	0.005	0.006	0.33	0.006	0.008	0.77	0.010	0.016	5.19
	SMCO-BR	0.003	0.008	0.21	0.007	0.010	0.35	0.010	0.015	0.81	0.010	0.017	5.41
(II)	GenSA	0.020	0.054	1.71	0.031	0.034	4.35	0.032	0.038	15.69	0.010	0.018	89.93
	SA	0.069	0.076	0.06	0.073	0.085	0.07	0.108	0.124	0.14	0.114	0.124	0.34
	DEoptim	0.013	0.054	0.30	0.036	0.038	0.76	0.045	0.050	2.78	0.056	0.064	13.18
	STOGO	0.052	0.052	3.49	0.068	0.068	8.18	0.080	0.080	28.43	0.082	0.082	182.28
	GA	0.050	0.058	0.22	0.036	0.049	0.25	0.054	0.074	0.38	0.059	0.070	0.65
	PSO	0.040	0.056	0.65	0.034	0.038	0.90	0.038	0.046	1.64	0.017	0.026	4.32
Empirical welfare maximization: $\max_{\beta} f_{EW}(\beta)$													
Best value		1,541.032			2,061.409			2,649.911			3,825.49		
(I)	GD	752	1,448	0.12	1,288	1,666	0.16	1,827	2,058	0.24	2,934	3,067	0.60
	SignGD	752	1,448	0.13	1,288	1,666	0.17	1,827	2,058	0.25	2,934	3,067	0.60
	ADAM	752	1,448	0.13	1,288	1,666	0.16	1,827	2,058	0.25	2,934	3,067	0.62
	SPSA	752	1,448	0.11	1,305	1,711	0.14	1,848	2,114	0.21	2,920	3,063	0.69
	L-BFGS	721	1,448	0.12	1,199	1,621	0.18	1,799	2,050	0.28	2,927	3,067	0.99
	BOBYQA	398	955	0.13	846	1,267	0.20	1,365	1,622	0.40	2,338	2,599	1.51
	SMCO	207	301	1.50	195	310	3.41	271	379	9.49	428	498	78.37
	SMCO-R	128	230	1.51	69	141	3.41	98	171	9.52	158	233	78.59
	SMCO-BR	140	229	1.55	85	198	3.46	144	252	9.57	274	361	78.71
(II)	GenSA	104	150	13.37	220	368	32.37	639	918	91.25	1,586	1,762	455.24
	SA	214	318	0.41	431	638	0.56	881	1,268	0.85	1,939	2,398	1.57
	DEoptim	116	161	2.37	233	279	6.11	640	723	17.54	1,722	1,805	87.03
	STOGO	134	134	27.94	470	470	66.33	851	851	181.67	2,096	2,096	891.62
	GA	160	238	1.08	288	453	1.36	689	973	1.90	1,685	1,831	3.69
	PSO	108	192	3.60	138	272	5.26	297	433	8.26	848	1,051	21.52

Based on 500/ $d = 100, 50, 25, 10$ reps. SMCO/R-BR and Group (I) algorithms run from $\sqrt{d} \approx 2, 3, 4, 7$ uniformly drawn starting points in parallel. Run on high-performance cluster with 8 cores on 1 node.

simulate $\epsilon_i \sim \mathcal{N}(0, d)$ and set $N = 500$. The variances of $X_{i,1}$ and ϵ_i are set equal to d so that the effects from $X_{i,1}$ and ϵ_i will not be dominated by those from $X_{i,-1} \in \mathbb{R}^d$ as the dimension d increases. We consider $d = 5, 10, 20$, and 50, and constrain the parameter search in the rectangle $[-20, 20]^d$.

The second randomly generated test function we consider arises from empirical welfare maximization as introduced in ref. 33, which has inspired a very active line of related research in economics on optimal treatment assignment and policy evaluation. Specifically, the empirical welfare function, say, under linear treatment assignment, is defined in ref. 33 as $f_{EW}(\beta) := \frac{1}{N} \sum_{i=1}^N \left(\frac{D_i}{p(X_i)} - \frac{1-D_i}{1-p(X_i)} \right) Y_i \mathbb{I} \left\{ \beta_1 + X_i' \beta_{-1} \geq 0 \right\}$ where $(D_i, X_i, Y_i)_{i=1}^N$ is a random sample with D_i being a binary treatment status variable with (known/chosen) propensity scores $p(X_i) := \mathbb{E}[D_i | X_i]$, $X_i \in \mathbb{R}^{d-1}$ being a vector of covariates, and Y_i is a scalar-valued outcome variable. The optimal linear treatment assignment rule is then obtained by solving the optimization problem $\max_{\beta} f_{EW}(\beta)$. Similarly to the maximum

score criterion function, f_{EW} also features the discontinuous indicator function that makes the optimization problem challenging. We also investigated the performance of the SMCO algorithms in solving empirical welfare maximization problems. Specifically, we build the test function f_{MS} based on a combination of real and simulated data. The real data are constructed based on the replication package of ref. 33, which is constructed from the Job Training Partnership Act (JTPA) data on job training programs. The specific dataset we use contains $N = 9,223$ observations on the binary treatment status variable D_i , the outcome variable Y_i (postprogram earning), and two covariates: preprogram earning (X_{i1}) and education (X_{i2}). The propensity score is known to be constant at $p(X_i) = 2/3$. Following ref. 33, we set $X_{i3} = X_{i2}^2$ and $X_{i4} = X_{i2}^3$ to be quadratic and cubic transformation of education. We then standardize $X_{i1}, \dots, X_{i4}$ by demeaning it and rescaling its SD to 1. We also augment the data by simulating additional covariates $X_{i5}, \dots, X_{i,d-1} \sim \mathcal{N}(0, 1)$ for higher dimensions $d \geq 6$. Again, we consider $d = 5, 10, 20$, and 50, and constrain the parameter space to $[-20, 20]^d$.

Table 3 reports the performance of our SMCO algorithms, along with the competing algorithms, in maximizing the two randomly generated test functions f_{MS} and F_{EW} as described above. First, all gradient-based local algorithms perform significantly worse than our SMCO algorithms and the global algorithms: This is expected given that the gradients of f_{MS} and F_{EW} are both degenerate almost everywhere given that the parameter enters through the indicator function. Second, our SMCO algorithms (especially SMCO-R/BR) perform robustly well across almost all the configurations in Table 3: In fact, SMCO-R/BR yields the best results among all algorithms except for one configuration (f_{EW} with $d = 5$).

SI Appendix, Appendix E contains supplemental numerical results on maximum score estimation and empirical welfare maximization up to $d = 100$, including replication of Table 3 using 10 replications only, as well as supplemental robustness checks of maximum score estimation with Cauchy errors and different bound buffer parameters of SMCO-R/BR.

3.5. Additional Numerical Experiments. We have conducted many more numerical experiments to check the wide applicability of our SMCO algorithms. See SI Appendix, Appendix F for results on two additional random test functions: one based on conditional moment inequalities in a dynamic binary choice model, the other based on mean squared in-class approximation error for ReLU neural networks. See SI Appendix, Appendix G for results on additional test functions in statistics, such as Cauchy log-likelihood, ReLU neural network out of sample predictions, ReLU neural network out of sample misclassification problems. This section contains statistical performance measurements such as “out-of-sample” MSEs and the “out-of-sample” misclassification error rate. These are different from but complementary to the optimization performance measurements we focus on in the main text. We have also tested many other deterministic test functions, such as Bukin No.6, Damavandi, Dropwave, Easom, EggHolder, JennrichSampson, McCormick, Mishra No.6, Qing, Rosenbrock, Schwefel, Shubert, Six-Hump Camel, Trefethen, and Zakharov functions, with dimensions up to 200 (when applicable). The main findings from all these exercises are that, our SMCO algorithms, especially SMCO-R and SMCO-BR, perform very well across all these different configurations, essentially without any change in the hyperparameters other than the number of starting points used. In fact, this observation suggests that the suitability of the SMCO algorithms for global optimization goes well beyond the theoretical guarantees we provide in Section 2.

4. Conclusion

Based on the strategic law of large numbers we establish, this paper proposes a unified two-armed decision model for black-box optimization. We transform the difficult problem of global optimization over a bounded rectangle region to optimal strategy formation of multiple two-armed decision models in infinite policy sets. We propose a class of Strategic Monte Carlo Optimization (SMCO) algorithms, for which we establish local and global convergence results. We also show via an extensive array of numerical experiments that our proposed SMCO algorithms perform consistently well across different complex multimodal test functions and different dimensions (up to $d = 1,000$), often outperforming a portfolio of state-of-the-art existing algorithms, even though our SMCO algorithms are currently not coded in C++. Our work opens the door to a promising area of future research that explores and improves the theoretical properties as well as the practical implementation of optimization algorithms under our StLLN-based framework.

Data, Materials, and Software Availability. The Job Training Partnership Act (JTPA) data used in Section 3 can be constructed based on the replication package of ref. 33, which is publicly available at <https://doi.org/10.3982/ECTA13288>. The R code used to generate the numerical studies in Section 3 and the Appendix are freely available for academic use on Github at <https://github.com/wayne-y-gao/SMCO> and <https://github.com/yanxiaodong128/SMCO>.

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